ADVANCED ML FOR PHYSICS

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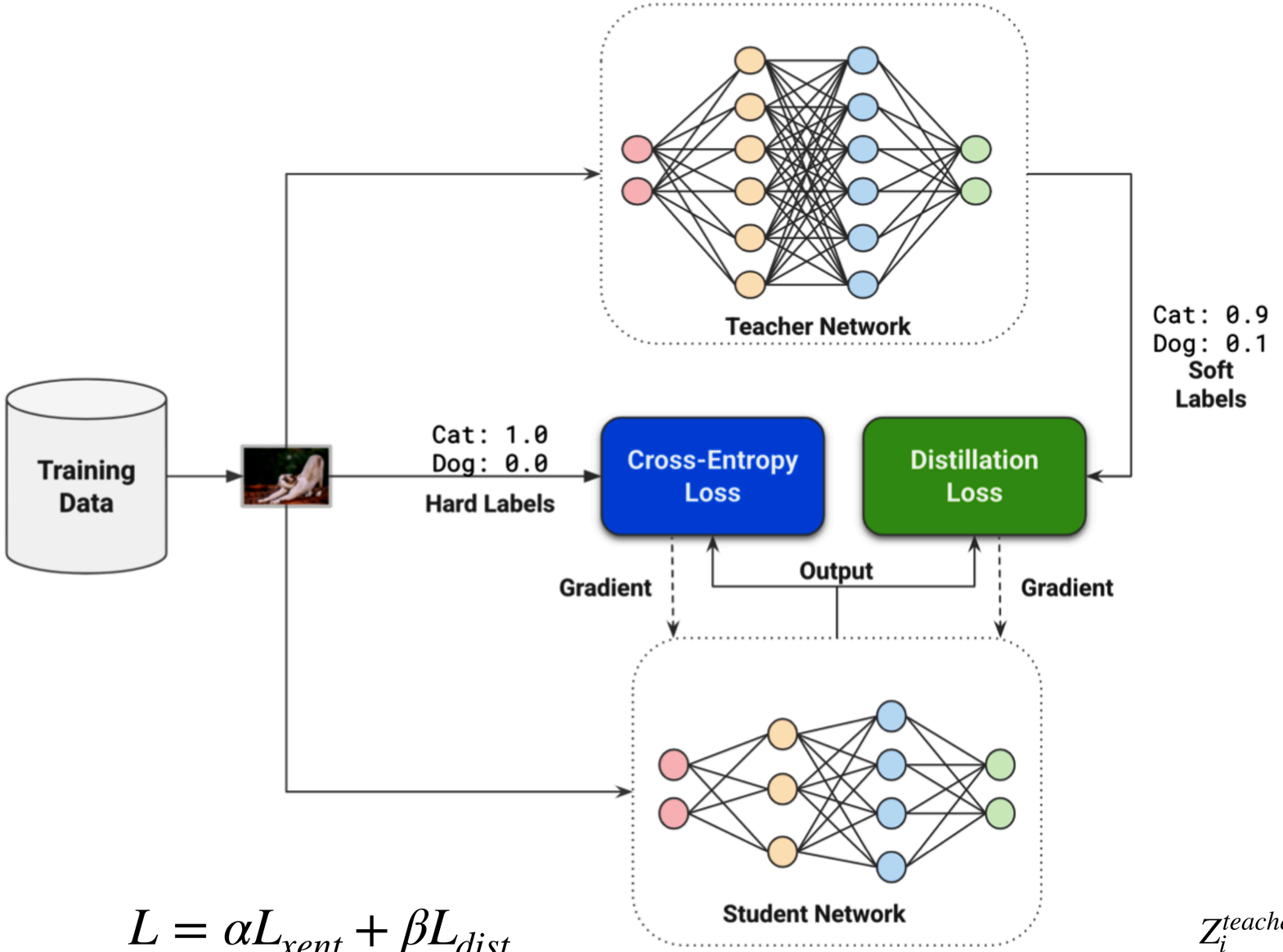
KNOWLEDGE TRANSFER VIA DISTILLATION

- Hinton and collaborators, in a seminal work from 2014 (arxiv:1503.02531) explored the possibility of teaching lightweight neural networks (students) to extract "dark knowledge" from a single complex teacher model or even from an ensemble of teachers
- Hinton's idea was to use a teacher model to generate soft labels for training data (somewhat similar to data augmentation where you create synthetic data to improve generalization ability of a neural network)
  - the soft labels replace the values of the ground-truth labels with the probabilities that the input belongs to each class, probabilities estimated by the teacher
  - during the training the student model can therefore learn both from the hard labels (ground truth) and from the soft ones produced by the teacher



- both teacher and student receive the same input
- the student is trained using the normal x-entropy loss crossed with the hard labels, and the distillation loss which uses the soft labels
- during training the teacher weights are frozen and therefore are not updated during backpropagation

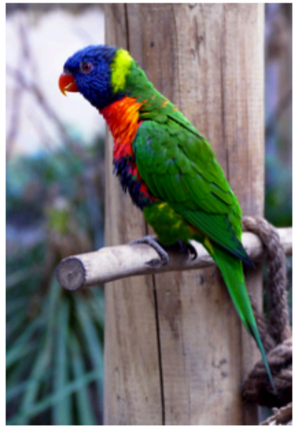
$$L = \alpha L_{xent} + \beta L_{dist}$$

$$L_{dist} = \text{x-entropy}(\hat{y}, y^{soft}; \mathbf{w})$$

$$y_i^{soft} = \frac{\exp \frac{Z_i^{teacher}}{T}}{\sum_j \exp \frac{Z_j^{teacher}}{T}}$$

T: temperature parameter which acts as a smoothing for the distribution of soft labels

intuitive explanation:

Labels	Images			
				
Hard[19]	[1, 0, 0, 0]	[0, 1, 0, 0]	[0, 0, 1, 0]	[0, 0, 0, 1]
Soft	[.80, .15, .03, .02]	[.15, .75, .05, .05]	[.03, .02, .85, .10]	[.05, .05, .20, .79]

- the teacher understand that a cat is more similar to a dog than a pigeon or a parrot, and that the latter are more similar to each other than to a cat or a dog
- distillation facilitate student's training by allowing to capture relationships between classes that are not represented in the hard labels of the training dataset

WHY DISTILLATION?

Operational Constraints

- Larger models learn “better”
- They get more benefit from having larger dataset
- Small model can be difficult to train (sensitive to initialization, LR, convergence, underfitting)

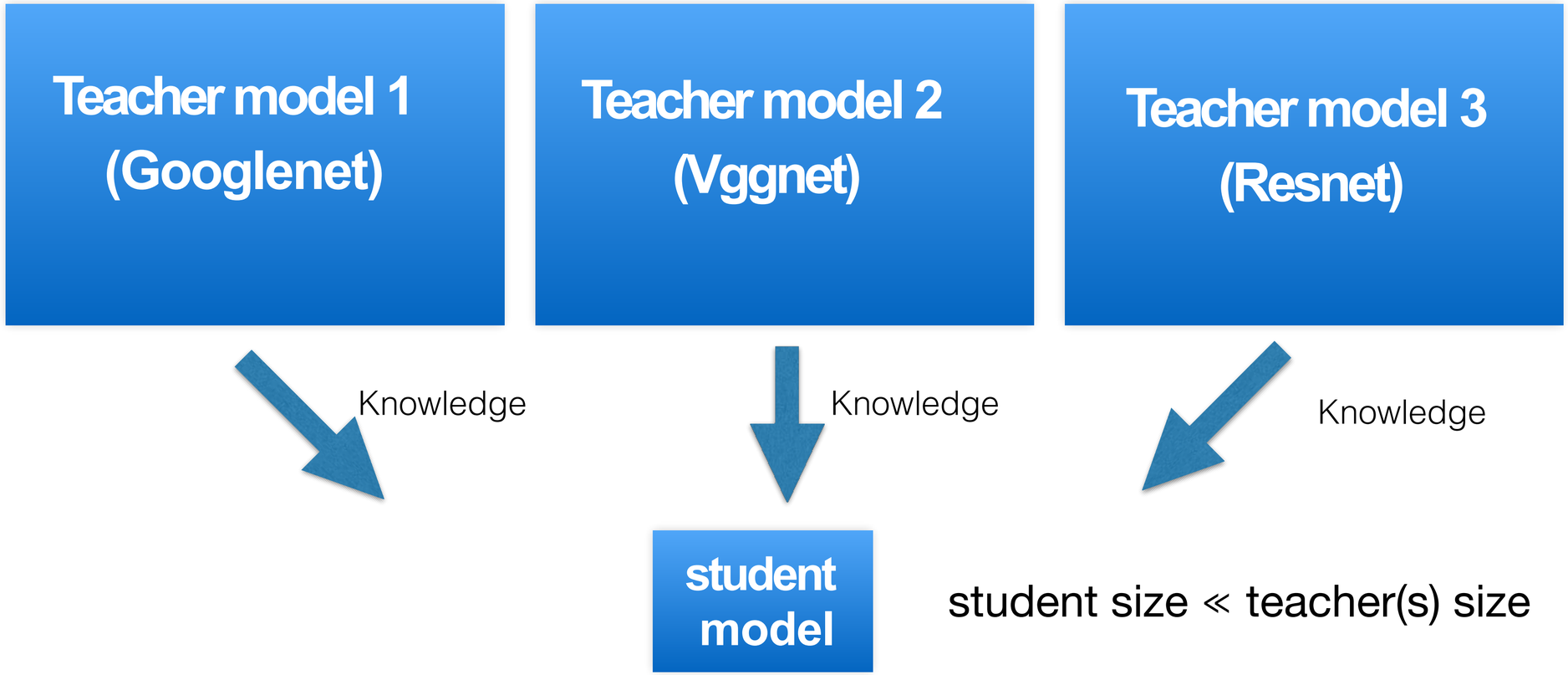
Data complexity

- Real-world data have complex topologies and features. Distillation can transfer structural and representational “know-how”

- **Performance improvement:** student (same or different size) learns from teacher’s pre-acquired knowledge to generalize better
- **Model compression:** smaller/faster student with comparable accuracy (example real-time applications)
- **Ensambale multiple models** w/o the cost of ensambling …

KNOWLEDGE DISTILLATION TO COMPRESS DNN MODELS

idea: transfer the knowledge learned from one or more DNNs of large size and expressive power, pre-trained for the same task, to a lightweight model possibly also compressed with the methods seen previously



student size ≪ teacher(s) size

Hinton et al, arxiv:1503.02531
Romero et al, arxiv:1412.6550

WHAT TO DISTILL?

Logits (soft targets)

- transfer output distributions/ scores. Acts like regularization and provides additional supervision (useful with few labels, smaller datasets)

- **Typical loss:** KL divergence or soft cross-entropy between teacher and student predictions

Embeddings (representations)

- transfer intermediate node/graph representations. Student "observes" the teacher's internal workflow beyond the final labels

- **Typical:** align hidden states for the same node (GNN)/layer (DNN/ CNN), or contrastive alignment across embedding spaces

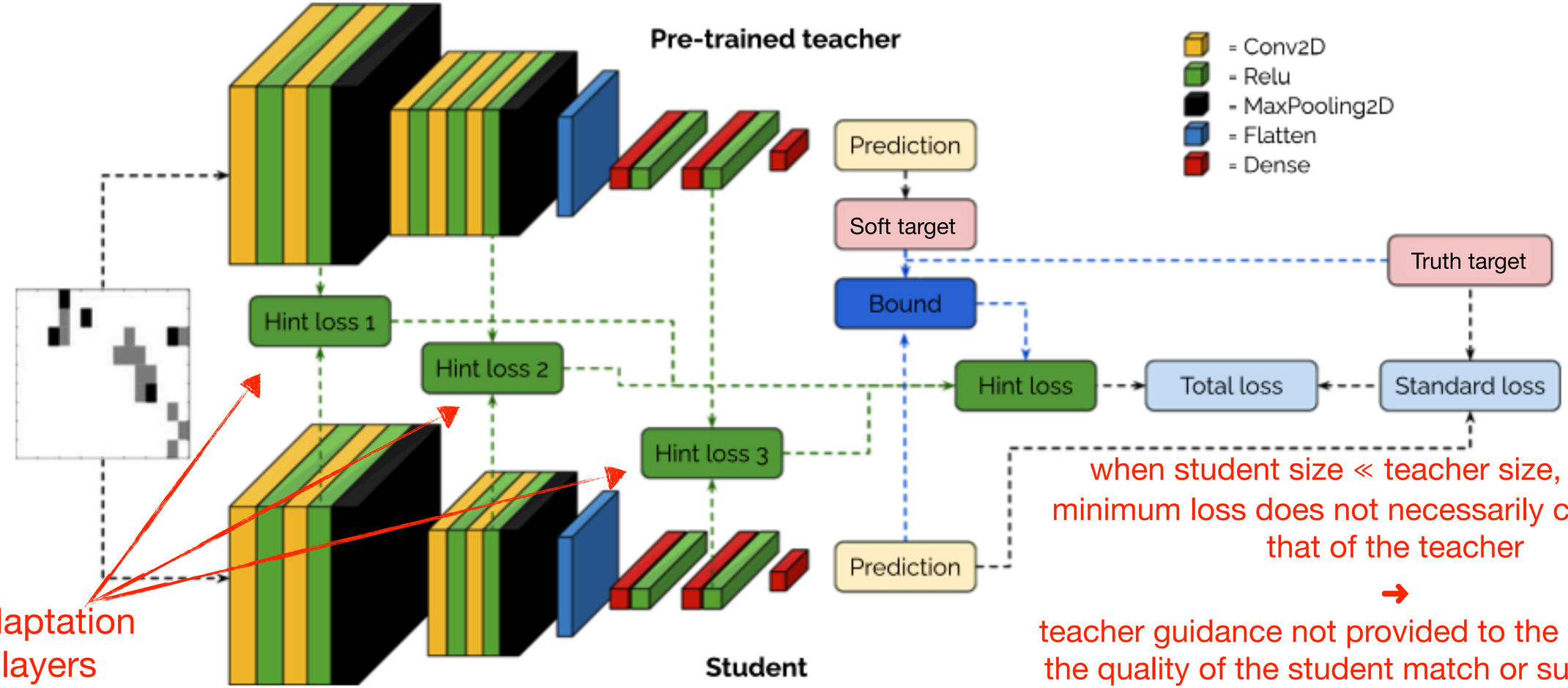
Structures

- transfer relational knowledge: ex. local neighborhoods and/or global topology signals in GNNs. Helps student retain connectivity semantics and multiscale structural cues

- **Local:** neighbor similarity vectors, discrepancy signals
- **Global:** meta-paths, kernels, positional/structural encodings

Knowledge Distillation on Graphs: A Survey, Tian et al, (2025) - doi:10.1145/3711121

ADVANCED KNOWLEDGE DISTILLATION



adaptation layers

when student size « teacher size, student's minimum loss does not necessarily coincide with that of the teacher

➜

teacher guidance not provided to the student once the quality of the student match or surpass that of the teacher with a certain margin

allows you to train very light models that would otherwise be impossible to make it to converge in the absence of the teacher

Romero et al, arxiv:1412.6550

S. Francescato, S.Giagu, F. Riti, G.Russo, L.Sabetta, F.Tortonesi, Eur. Phys. J. C (2021) 81:969

EXAMPLE OF "GUIDED" LOSS

- the teacher's guidance is provided to the student only until the student's performance falls short of that of the teacher by a certain margin, the margin is controlled by an appropriate hyper-parameter γ

total loss $L(y, y_S, y_T)$

$$= \begin{cases} \|y - y_S\|^2 + \sum_{i=1}^{3} \gamma_i H_i & \text{if } \|y - y_T\|^2 < \gamma \|y - y_S\|^2 \\ \|y - y_S\|^2 & \text{Otherwise} \end{cases}$$

$$H_i = \|A_i - T_i^H\|^2$$

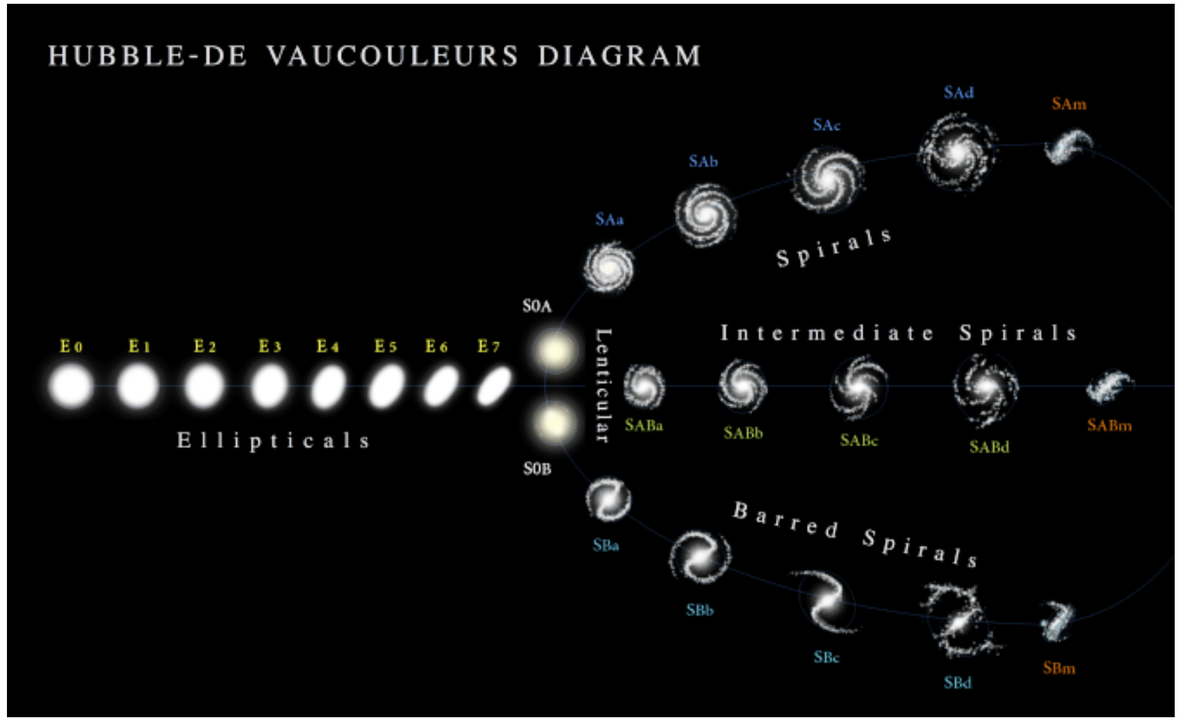
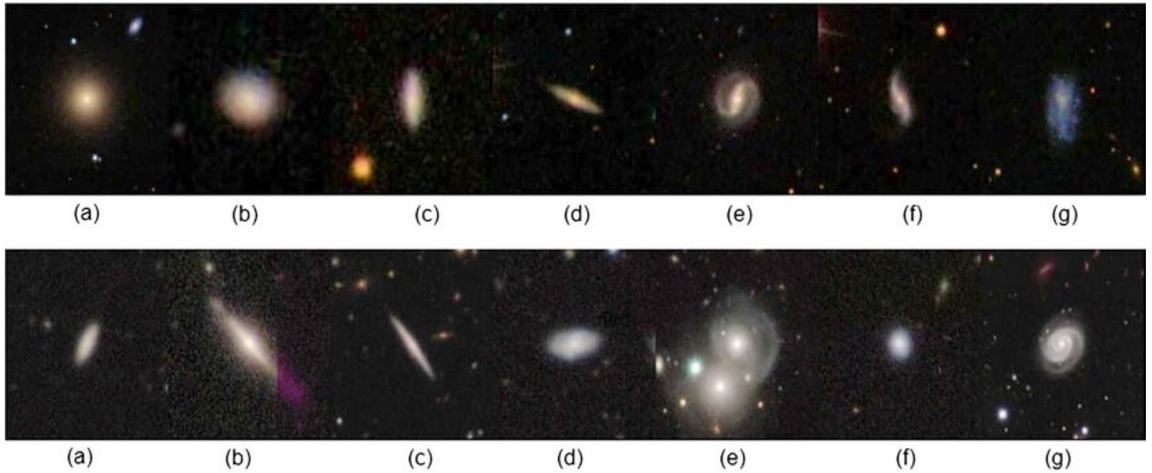
hint loss (adaptation layer)

y_S: student prediction

y_T: teacher prediction

γ: hyper-parameter to control the effect of teacher's guidance

$\|\cdot\|^2$: L2 norm

GALAXY MORPHOLOGICAL CLASSIFICATION

- Wikipedia
- Galaxy Zoo 2 dataset
  - Willett, Kyle W., “Galaxy Zoo 2: Images from Original Sample”, Monthly Notices of the Royal Astronomical Society, vol. 435, no. 4. Zenodo, pp. 2835–2860, Nov. 01, 2013. doi:10.5281/zenodo.3565489
  - H. Domínguez Sánchez, M. Huertas-Company, M. Bernardi, D. Tuccillo, and J. L. Fischer, ‘Improving galaxy morphologies for SDSS with Deep Learning’, Monthly Notices of the Royal Astronomical Society, vol. 476, pp. 3661–3676, Feb. 2018, doi:10.1093/mnras/sty338